

1. Basic Probability & Distributions

Probability Functions

- **PDF (Continuous):** A function $f(x)$ such that

$$P(a \leq X \leq b) = \int_a^b f(x) dx,$$
$$\text{with } \int_{-\infty}^{\infty} f(x) dx = 1.$$

- **PMF (Discrete):** A function $p(x)$ where

$$P(X = x) = p(x), \quad \sum_x p(x) = 1.$$

- **CDF:**

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$
$$(\text{or } \sum_{t \leq x} p(t)).$$

- **Joint CDF (two variables):**

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y).$$

Transformation of Variables

- **Change of Variables:** If $Y = g(X)$ is invertible, then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy}(g^{-1}(y)) \right|.$$

- **Probability Integral Transform (PIT):** If X is continuous with CDF F , then

$$F(X) \sim \text{Uniform}(0, 1).$$

2. Moments & Variance

Moments

- The k -th moment:

$$\mu_k = E[X^k] = \int_{-\infty}^{\infty} x^k f(x) dx$$
$$(\text{or } \sum x^k p(x) \text{ for discrete}).$$

- **Mean:** $E[X] = \mu_1$.

- **Variance:**

$$\text{Var}(X) = E[X^2] - (E[X])^2.$$

Sample Moments

$$\hat{\mu}_k = \frac{1}{n} \sum_{i=1}^n X_i^k.$$

3. Central Limit Theorem (CLT)

If $X_1, X_2, \dots, X_n$ are i.i.d. with $E[X] = \mu$ and $\text{Var}(X) = \sigma^2$, then for large n :

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

Standardization:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

4. Properties of Point Estimators

- **Unbiasedness:** $\hat{\theta}$ is unbiased if

$$E[\hat{\theta}] = \theta.$$

- **Mean Squared Error (MSE):**

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$
$$= \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2.$$

- **Consistency:**

$$\hat{\theta}_n \xrightarrow{p} \theta \quad (\text{as } n \rightarrow \infty).$$

- **Efficiency:** Relative efficiency of two estimators $\hat{\theta}_1$ and $\hat{\theta}_2$:

$$\text{Eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}.$$

5. Likelihood, Sufficient Statistics, and MVUE

- **Likelihood Function:**

$$L(\theta | x) = \prod_{i=1}^n f(x_i | \theta).$$

- **Log-Likelihood:**

$$\ell(\theta) = \log L(\theta | x) = \sum_{i=1}^n \log f(x_i | \theta).$$

- **Factorization Theorem:** $T(x)$ is sufficient for θ if

$$L(\theta | x) = g(T(x), \theta) h(x),$$

where $h(x)$ does not depend on θ .

- **Rao-Blackwell Theorem:** If $\hat{\theta}$ is unbiased and $T(x)$ is sufficient for θ , then

$$\hat{\theta}^* = E[\hat{\theta} | T(x)]$$

is the MVUE for θ (with lower variance than $\hat{\theta}$).

6. Method of Moments (MoM) Estimation

- **Procedure:**

1. Compute theoretical moments in terms of θ , e.g. $E[X] = \mu(\theta)$, $E[X^2] = \mu_2(\theta)$.
2. Compute sample moments: $\hat{\mu}_1 = \frac{1}{n} \sum_{i=1}^n X_i$, $\hat{\mu}_2 = \frac{1}{n} \sum_{i=1}^n X_i^2$, etc.
3. Equate and solve: $\mu(\theta) = \hat{\mu}_1$, etc.

- **Example: Uniform**(0, θ)

$$E[X] = \frac{\theta}{2} \implies \hat{\theta} = 2\bar{X}.$$

7. Maximum Likelihood Estimation (MLE)

- **Steps:**

1. Write the likelihood:

$$L(\theta | x) = \prod_{i=1}^n f(x_i | \theta).$$

2. Take the log-likelihood:

$$\ell(\theta) = \sum_{i=1}^n \log f(x_i | \theta).$$

3. Differentiate:

$$\frac{d\ell(\theta)}{d\theta} = 0.$$

Solve for θ .

- **Invariance:** If $\hat{\theta}$ is the MLE for θ , then for any function $g(\theta)$, the MLE for $g(\theta)$ is $g(\hat{\theta})$.

8. Fisher Information & Asymptotic Properties

- **Score Function:**

$$U(\theta) = \frac{\partial}{\partial \theta} \log L(\theta | x).$$

- **Fisher Information:**

$$I(\theta) = E\left[\left(\frac{\partial}{\partial \theta} \log L(\theta | x)\right)^2\right]$$
$$= -E\left[\frac{\partial^2}{\partial \theta^2} \log L(\theta | x)\right].$$

- **Asymptotic Normality:** For large n ,

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, I(\theta)^{-1}),$$

or equivalently,

$$\hat{\theta} \sim N\left(\theta, \frac{I(\theta)^{-1}}{n}\right).$$

9. Transformation of PDFs

- **Change of Variables:** If $Y = g(X)$ (invertible), then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy}(g^{-1}(y)) \right|.$$

10. Multivariate Distributions

- **Joint PDF:** $f_{X,Y}(x,y)$ describes the density at (x,y) .
- **Marginal PDF:**

$$f_X(x) = \int f_{X,Y}(x,y) dy.$$

- **Conditional PDF:**

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}.$$

11. Central Limit Theorem (CLT) Recap

If $X_1, \dots, X_n$ are i.i.d. with mean μ and variance σ^2 , then for large n :

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

Standardization:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

Extra Tips

- **Indicator functions:** $I_A(x) = 1$ if $x \in A$, else 0.
- **Log of product:** $\log\left(\prod_{i=1}^n a_i\right) = \sum_{i=1}^n \log(a_i)$.
- **Sufficient stats:** e.g. $\sum X_i, \max X_i$ for certain families.

Appendix: Common Distributions

Discrete Distributions

Y	Parameters	$\mathbf{p(y)}$	$\mathbf{F(y)}$	$\mathbf{E[Y]}$	$\mathbf{Var(Y)}$	$\mathbf{mgf(t)}$
Bernoulli (p)	$Y \in \{0,1\}, p \in (0,1)$	$p^y(1-p)^{1-y}$	step function	p	$p(1-p)$	$1-p+pe^t$
Binomial (n,p)	$Y \in \{0,\dots,n\}$	$\binom{n}{y}p^y(1-p)^{n-y}$	$\sum_{k=0}^y \binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$	$[1-p+pe^t]^n$
Poisson (λ)	$Y \in \{0,1,2,\dots\}, \lambda > 0$	$\frac{\lambda^y e^{-\lambda}}{y!}$	$\sum_{k=0}^y \frac{\lambda^k e^{-\lambda}}{k!}$	λ	λ	$\exp[\lambda(e^t-1)]$
Geometric (p)	$Y \in \{1,2,\dots\}$	$p(y) = p(1-p)^{y-1}$	$1-(1-p)^y$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-e^t(1-p)}$
Hypergeometric (N,r,n)	$Y \in \{\max(0,n-(N-r)),\dots,\min(n,r)\}$	$\frac{\binom{r}{y}\binom{N-r}{n-y}}{\binom{N}{n}}$	cumulative sum	$n \frac{r}{N}$	$n \frac{r}{N} \left(1-\frac{r}{N}\right) \frac{N-n}{N-1}$	no simple mgf

Continuous Distributions

Y	Parameters	$\mathbf{f(y)}$	$\mathbf{F(y)}$	$\mathbf{E[Y]}$	$\mathbf{Var(Y)}$	$\mathbf{mgf(t)}$
Uniform ($0,\theta$)	$\theta > 0$	$\frac{1}{\theta}, 0 < y < \theta$	$\frac{y}{\theta}, 0 < y < \theta$	$\frac{\theta}{2}$	$\frac{\theta^2}{12}$	$\frac{e^{t\theta}-1}{t\theta}$
Exponential (β)	$\beta > 0$	$\frac{1}{\beta}e^{-y/\beta}, y > 0$	$1-e^{-y/\beta}$	β	β^2	$\frac{1}{1-\frac{t}{\beta}} \quad (t < \frac{1}{\beta})$
Gamma (α,β)	$\alpha > 0, \beta > 0$	$\frac{1}{\Gamma(\alpha)\beta^\alpha}y^{\alpha-1}e^{-y/\beta}$	$\frac{\gamma(\alpha,y/\beta)}{\Gamma(\alpha)}$	$\alpha\beta$	$\alpha\beta^2$	$(1-\beta t)^{-\alpha} \quad (t < \frac{1}{\beta})$
Normal (μ,σ^2)	$\sigma^2 > 0$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right]$	$\Phi\left(\frac{y-\mu}{\sigma}\right)$	μ	σ^2	$\exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$